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NIH Websites Are Changing

- Get the Latest Updates and FAQs on the transition
- Per Executive Order 14434, federal agencies are now using "Super Intelligence" (SI) in place of "Artificial Intelligence" (AI). You may see both terms across NIH content during this transition.

NM_000298.6(PKLR):c.1457G>A (p.Arg486G...

Cite Follow Print Download

Germline

Classification

Uncertain significance
Uncertain significance (2)
2 out of 3 submissions contributed to this classification

Somatic

No data submitted for somatic clinical impact

tic

tted for oncogenicity

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Start Survey

No Thanks

Functional data are available for this v

Variant Details

Variant details

Identifiers: NM_000298.6(PKLR):c.1457G>A (p.Arg486Gln)

Variation ID: 873616 Accession: VCV000873616.7

Type and length: single nucleotide variant, 1 bp **Location:**

Cytogenetic: 1q22 1: 155291917 (GRCh38) [NCBI UCSC] 1: 155261708 (GRCh37) [NCBI UCSC] **Timeline in ClinVar:**

	First in ClinVar	Last submission	Last evaluated
Germline	May 31, 2020	Sep 6, 2025	Apr 4, 2023

HGVS:

Nucleotide	Protein	Molecular consequence
NM_000298.6:c.1457G>A MANE SELECT	NP_000289.1:p.Arg486Gln	missense
NM_181871.4:c.1364G>A	NP_870986.1:p.Arg455Gln	missense
NC_000001.11:g.155291917C>T		

... more HGVS

Protein change: R486Q, R455Q **Other names:**

NC_000001.10:g.155261708C>T **Canonical SPDI:** NC_000001.11:155291916:C:T

Global minor allele frequency (GMAF) : -**Allele frequency :**

The Genome Aggregation Database (gnomAD) 0.00001

Trans-Omics for Precision Medicine (TOPMed) 0.00001

The Genome Aggregation Database (gnomAD), exomes 0.00003

The Genome Aggregation Database (gnomAD), exomes 0.00004

Exome Aggregation Consortium (ExAC) 0.00007

Links : ClinGen: CA1143991
dbSNP: rs768210809
VarSome

Genes

Feedback

Gene	OMIM	ClinGen Gene Dosage Sensitivity Curation		Variation Viewer	Related variants	
		HI score	TS score		Within gene	All
PKLR		-	-	GRCh38 GRCh38 GRCh37	482	513

Conditions - Germline

Condition	Classification (# of submissions)	Review status	Last evaluated	Variation/condition record
Pyruvate kinase deficiency of red cells	Uncertain significance (1)	★☆☆☆	Apr 27, 2017	RCV001095924.4
not provided	Uncertain significance (1)	★☆☆☆	Apr 4, 2023	RCV003490065.3

Submissions - Germline

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows
Uncertain significance (Apr 27, 2017) C Contributing to aggregate classification	★☆☆☆ (ICSL Variant Classification Criteria 13 December 2019)	Pyruvate kinase deficiency of red cells	Illumina Laboratory Services, Illumina 252103.1 31, 2020 31, 2020	
Uncertain significance (Apr 04, 2023) C Contributing to aggregate classification	★☆☆☆ (ACMG Guidelines, 2015)	not provided	Revvity Omics, Revvity Accession: SCV004236096.2 First in ClinVar: Feb 04, 2024 Last updated: Sep 06, 2025	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: unknown

Comment:
This variant was observed as part of a p
applicable). No publications were found ba
is classified as a variant of unknown signif
rch was performed for the gene, cDNA change, and amino acid change (where
allow this variant to be ruled in or out of causing disease. Therefore, this variant

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

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Citations for germline classification of this variant

There are no citations for germline classification of this variant in ClinVar. If you know of citations for this variation, please consider submitting that information to ClinVar.

Submissions - Functional Data

functionally abnormal -- retained intron

assessed by **RNAseq**

Result: read abundance - intron inclusion (p=0.0112)

Modifies an exonic cryptic acceptor site at NC_000001.10:g.155261707 from -13.04 to 1.68 bits. Significant

Dr. Peter K. Rogan Lab, Western University

Accession: SCV006596601.1 Submitted: 2025-10-17

Assay description:

p-values: read abundance - intron inclusion (4 reads, p=0.0112). Computations were originally performed using genomic coordinates. This variant was mapped to an appropriate current MANE database designation or, if not present in this source, to the longest relevant transcript in the Biocommons Universal Transcript Archive using the hgvs Python package. The original genomic coordinate was based on GRCh37 and was submitted as an "Alternate designation". Modifications of Figure 2 of PMID:31275557 for ClinVar: a) "strongly corroborating Veridical p-value" refers to "junction-spanning" evidence types; b) add the text "and natural site < 300 nucleotides away?" to "Ri,final > nearest natural site Ri"; c) "Aberrant" and "Likely aberrant" classifications are now both defined as "Functionally abnormal" (analogous to ACMG/AMP reporting); d) "Allele-specific alternative splicing" classification now defined as "Function uncertain" (based on results for similar variants in PMID: 32211018); e) average heterozygosity based-criteria no longer contribute to classification.

Citation: PubMed: 31275557

In the sample (TCGA-FG-7636), the counts of splice junction spanning reads and abundance of the transcript adjacent to this mutation were compared with RNASeq data from matched tissues and tumors lacking the same mutation using Veridical (PMID:24741438)

- Disease context: Glioma susceptibility 1
- Transcript, protein change: NM_000298.6:c.1457G>A, R486Q
- Molecular phenotype measured: splicing
- Cell line: TCGA-FG-7636
- Tissue: Brain Lower Grade Glioma (LGG)
- Collection method: in vitro
- Species: human
- Number of controls: 389

Text-mined citations for rs768210809

LitVar 2

These citations are identified by LitVar using the rs number, so they may include citations for more than one variant at this location. Please review the LitVar results ca

Record last updated Apr 13, 2026

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